

Enhancing Strabismus Diagnosis from Detection to Classification with Deep Learning

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Abstract—Accurate and early diagnosis of strabismus, a disorder characterized by ocular misalignment, is essential to prevent long-term visual impairment. Traditional diagnostic approaches heavily rely on clinical expertise, often introducing subjectivity and inconsistency. In this study, we propose a deep learning-based framework for automated strabismus classification using four state-of-the-art convolutional neural networks—EfficientNet-B7, ResNet-50, AlexNet, and VGGNet-19. The models are trained on a diverse open-source dataset to classify strabismus into five categories: Esotropia, Exotropia, Hypertropia, Hypotropia, and Normal eye alignment. Our extensive experiments demonstrate that EfficientNet-B7 outperforms other models, achieving superior accuracy and generalization across evaluation metrics, including precision, recall, and F1-score. The proposed AI-driven approach offers an objective, scalable, and consistent alternative to manual diagnosis, significantly enhancing clinical decision-making. This study underscores the transformative potential of deep learning in ophthalmology, paving the way for automated, high-accuracy strabismus screening in real-world clinical settings. Future work will explore real-time implementation and integration of additional clinical parameters to further optimize performance and usability.

Index Terms—Strabismus Classification, Deep Learning, Ophthalmology, Medical image

I. INTRODUCTION

Strabismus, also called ocular misalignment, is a visual disorder affecting people of all age groups. It may result in binocular vision dysfunction (BVD) [1], disorders in depth perception, eye coordination problems, and, in extreme cases, amblyopia. Patients affected with strabismus often experience difficulties in day-to-day activities like reading, driving, and sports due to impaired visual coordination. Social and psychological effects, such as difficulty maintaining eye contact and decreased self-esteem, also affect quality of life. Therefore, a timely and accurate diagnosis is essential for timely intervention, as untreated strabismus can lead to lasting visual disabilities and psychosocial problems.

Conservative assessment methods such as the Hirschberg test [2], cover tests, and prism evaluations, which depend on traditional diagnostic methods, are subjective and necessitate specialized and dedicated expertise. However, these methods

lack automated techniques and quantitative accuracy and require good resources, further restricting their scalability in remote and resource-constrained environments.

With advancements in artificial intelligence (AI) [3] and deep learning-based domains [4], computer vision-based diagnostic tools have demonstrated significant potential in ophthalmology. Current AI-driven models dedicated to strabismus detection concentrate mainly on classification. They mainly focus in differentiating between normal and misaligned eye positions. Although these approaches have yielded hopeful outcomes, they fail to provide quantitative measurement abilities. One cannot rely on these methods to plan clinical treatments and decisions further. There is an acute requirement for a cohesive system categorizing strabismus subtypes and delivering exact forecasts.

This research presents a framework for thoroughly diagnosing strabismus based on deep learning [5], using several advanced models to achieve high precision. Among these, the most precise classification model is portrayed by EfficientNetB7, with diagnostic performance experiencing a notable enhancement. In addition to EfficientNetB7 [6], we incorporate ResNet-50 [7], VGG-19 [8], and AlexNet [9] to compare and enhance classification accuracy. Using open-source datasets, the framework categorizes strabismus into five types — esotropia, exotropia, hypertropia, hypotropia, and normal alignment. A robust, automated, and quantitative tool for diagnosing strabismus is aimed to be provided to ophthalmologists by integrating multiple classification models in our framework.

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II. LITERATURE REVIEW

Conventional methods for diagnosing strabismus, including the Hirschberg test, cover test, and prism measurement, rely on

the expertise and direct visual assessment of healthcare professionals [10]. Recent advancements in CNN-based models have demonstrated remarkable accuracy in detecting strabismus. Zheng et al. (2021) created a deep learning model for the detection of horizontal strabismus, reaching an AUC of 0.99, surpassing resident ophthalmologists.

Deep learning-driven classification systems have greatly progressed automated strabismus diagnosis. Robust performance in eye region segmentation and classification for telemedicine applications has been demonstrated by previous works, such as RF-CNN by Jiewei Lu (2018) [11]. Building upon these advancements, our study utilizes a multi-class classification method to differentiate between Esotropia, Exotropia, Hypertropia, Hypotropia, and Normal alignment through the application of various deep - learning frameworks.

To enhance classification accuracy further, CNN architectures such as EfficientNet have been utilized to concentrate on feature extraction. Arif et al. (2023) demonstrated that EfficientNet-B0 enhances ophthalmic image classification, particularly for fundus-based eye disease detection [12]. Driven by this, our research includes EfficientNet-B7, which enhances strabismus detection precision by minimizing unnecessary facial variations and refining feature extraction.

A multi-class classification approach is employed by our research to distinguish between Esotropia, Exotropia, Hypertropia, Hypotropia, and Normal alignment using various deep learning architectures. By utilizing models such as AlexNet, VGG19, ResNet50, and EfficientNetB7, we proficiently extract and categorize unique ocular characteristics essential for strabismus identification.

As advancements occur in AI-driven ophthalmic diagnostics, maintaining clinical reliability and generalizability continues to present a challenge [13]. Our approach seeks to deliver a precise and automated classification system, minimizing dependence on manual assessments and improving early identification and treatment preparation. Future work may focus on further refining classification models to achieve greater robustness and clinical applicability.

III. METHODOLOGY

Our deep learning-oriented framework for strabismus diagnosis focuses on classification using multiple deep learning architectures, provides an efficient diagnostic tool for ophthalmologists. The process is shown in Fig. 1.

A. Data Collection

The dataset was sourced from publicly available open-source platforms, including Kaggle, GitHub, and other online-available repositories. The collected images were manually classified into five categories: Esotropia, Exotropia, Hypertropia, Hypotropia, and Normal alignment. This ensured a high-quality dataset suitable for training and evaluating the classification models.

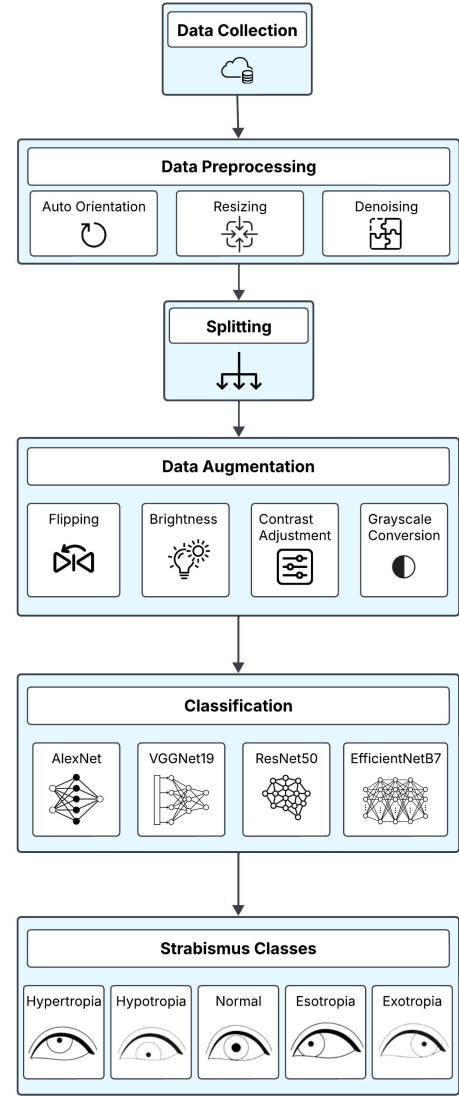


Fig. 1. Pipeline of the Proposed Model

B. Data Preprocessing

To standardize the dataset and enhance model performance, various preprocessing steps were applied to ensure consistency, enhance feature extraction, and boost training efficiency.

1) *Auto-orientation*: Image alignment was automatically corrected to maintain a consistent orientation across the dataset, minimizing discrepancies caused by camera positioning or subject posture.

2) *Resizing*: All images were resized to 227×227 pixels to match with the input requirements of the models. This ensured uniformity while preserving essential visual details necessary for accurate classification.

3) *Denoising*: To improve image clarity and minimize artifacts, a denoising procedure was utilized, further enhancing the dataset for better classification precision.

C. Splitting

The preprocessed data was divided into training, validation, and test sets using a 70:15:15 ratio, providing a robust evaluation framework.

D. Data Augmentation

A dedicated data augmentation and expansion process was applied to the training set to enhance data generalization. This approach also helped reduce excessive adaptation and mitigate overfitting. This was done to enhance the dataset's generalization and also helped control and reduce model learning and overfitting. Transformation techniques applied controlled modifications to the data, enhancing the model's adaptability and preventing it from memorizing specific patterns. These enhancements ensured that the model developed robust and generalizable properties. The applied augmentation methods included:

1) *Flipping*: Left and right gaze variations were considered by applying horizontal mirroring to images. This exposed the model to various facial orientations, enabling it to learn and adapt to different gaze directions more effectively.

2) *Brightness*: To enhance model strength against varying lighting conditions, brightness levels were modified between -20% and 20%. This adjustment prevented the model from developing a bias toward specific lighting conditions, significantly enhancing its performance across various illumination settings and environments.

3) *Grayscale Conversion*: A subset of images was converted to grayscale to minimize dependence on color-based features and enhance feature robustness. This encouraged the model to prioritize structural and texture-based patterns, enhancing its ability to generalize effectively.

4) *Brightness*: To enhance model strength against varying lighting conditions, brightness levels were modified between -20% and 20%. This adjustment prevented the model from being biased toward specific lighting levels. This further enhanced the model's performance across various illumination conditions and environments.

5) *Contrast Adjustment*: To further enhance the model's robust behavior against more varying lighting conditions, contrast levels were adjusted between -20% and 20%. Doing this prevented the model from becoming biased and influenced toward certain contrast levels. This process also improved model and data generalization. A balancing feature visibility technique also ensured important details remained intact and detectable in different high and low contrast settings.

E. Classification

Four deep learning models were utilized to accurately classify different types of strabismus. Each model was pre-trained on large-scale image datasets to leverage their learned feature representations effectively.

1) *AlexNet*: The model based on AlexNet was used to examine feature extraction abilities and comprehend the framework in which the model operates. The architecture incorporated DepthwiseConv2D layers and integrated techniques

such as adaptive learning rate scheduling and early stopping to enhance training efficiency. Additionally, the model employed Mixed Precision Training to enhance computational efficiency and made use of the AdamW optimizer for better weight regularization and convergence.

2) *VGGNet19*: The VGGNet-19 model was trained for categorization. Key components such as Flatten layers, Batch Normalization, and Dropout were incorporated by this architecture to enhance generalization and stability during training. Furthermore, adjustment of fully connected layers was carried out to enhance performance. A progressive unfreezing approach was applied, where deeper layers were gradually unfrozen to improve task-specific feature extraction. Meanwhile, the outer layers remained frozen to retain previously learned feature representations, ensuring a balance between stability and adaptability.

3) *ResNet50*: ResNet-50 was employed for feature analysis and strabismus image classification, succeeding the VGGNet-19 model. The base model was initialized using a pre-trained ResNet-50 architecture, with the last 75 layers unfrozen to facilitate domain-specific feature learning. Essential components, including Conv2D layers, Global Average Pooling, Dropout, and Layer Normalization, were incorporated into the model architecture. Additionally, fully connected layers with ReLU activation were incorporated to enhance non-linearity and improve classification accuracy. Softmax activation was used for the final classification output to ensure probability-based predictions.

4) *EfficientNetB7*: EfficientNet-B7 was selected as the final model for multi-class classification, as it outperformed all other evaluated architectures. To improve training stability and generalization, the model integrated Conv2D layers, Global Average Pooling, Weight Decay regularization, and an adaptive learning rate optimizer. To enable task-specific feature extraction, the last 30 layers were unfrozen, allowing the model to refine its representations based on the dataset.

F. Evaluation Metrics

In this section, we have calculated the parameters which are used to analyze the performance of the classification based models. The following metrics are commonly used to evaluate the performance of classification models:

1) *Accuracy*: It is represented in Eq. 1 by the given formula.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

where:

- *TP* is True-Positives i.e. Number of correct predictions of the positive class.
- *TN* is True-Negatives i.e. Number of correct predictions of the negative class.
- *FP* is False-Positives i.e. Number of incorrect predictions of the positive class.
- *FN* is False-Negatives i.e. Number of incorrect predictions of the negative class.

2) *Precision*: It is depicted in Eq 2 by it's formula.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

where:

- *TP* is True-Positives i.e. Number of correct predictions of the positive class.
- *FP* is False-Positives i.e. Number of incorrect predictions of the positive class.

3) *Recall (or Sensitivity)*: It is represented in Eq. 3 by its formula.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

where:

- *TP* is True-Positives i.e. Number of correct predictions of the positive class.
- *FN* is False-Negatives i.e. Number of incorrect predictions of the negative class.

4) *F1 Score*: It is shown in Eq. 4 by it's formula.

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

where:

- $\text{Precision} = \frac{TP}{TP + FP}$
- $\text{Recall} = \frac{TP}{TP + FN}$

IV. SIMULATION RESULTS

In this study, the classification of strabismus was focused using deep learning. The proposed model successfully performed the multiclass classification of strabismus i.e. Esotropia, Exotropia, Hypertropia, Hypotropia and Normal using different and popular computer vision architectures like EfficientNet B7, ResNet 50, VGGNet 19, and AlexNet out of which EfficientNet B7 performed the best, achieving the highest accuracy of 84.00% as shown in Table.I in detecting and classifying different types of strabismus which are represented in confusion matrices as shown in Fig. 2, 3, 4 and 5.

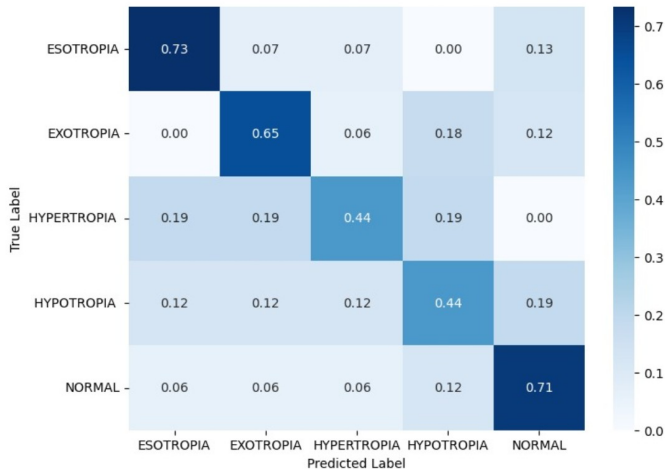


Fig. 2. Confusion matrix using AlexNet Model

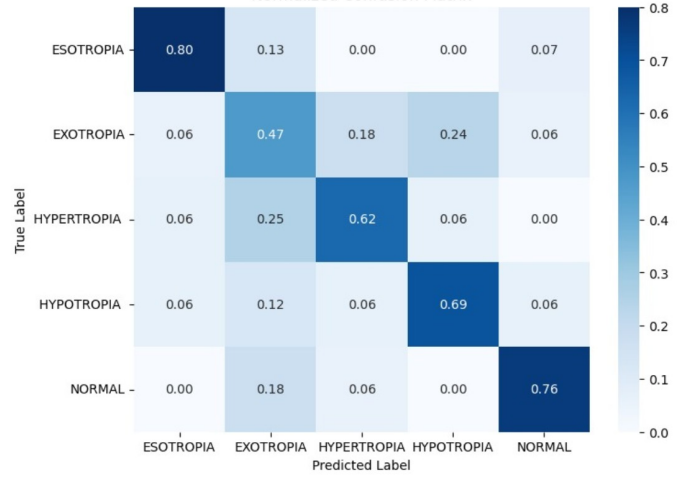


Fig. 3. Confusion matrix using VGGNet Model

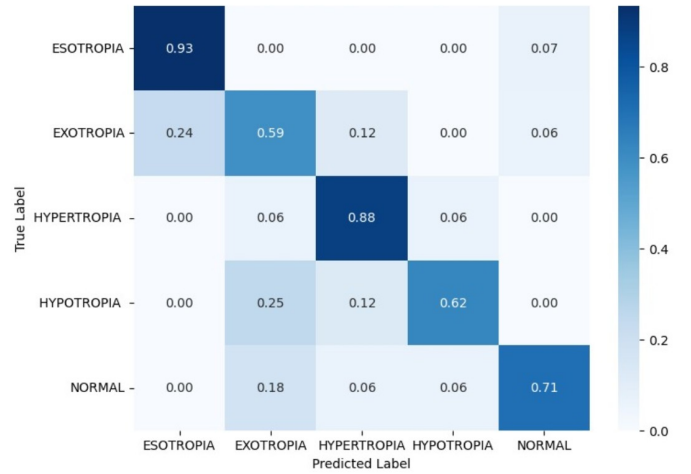


Fig. 4. Confusion matrix using ResNet-50 Model

These confusion matrices typically gives us a close analysis of models performance and key performance metrics such as accuracy, precision, recall and F1-score were computed as shown in Fig. 6, 7, 8 and 9. The simulation results defines the efficacy of using deep learning models for strabismus classification, demonstrating promising clinical applicability.

V. CONCLUSION

In this research, a method for classifying strabismus based on deep learning was suggested, utilizing various deep learning architectures. The models exhibited remarkable accuracy in multi-class classification tasks, successfully differentiating among the different types of strabismus.

This initiative supports the development of an automated AI-powered strabismus diagnosis system, reducing reliance on manual assessments and enhancing the efficiency of early identification. By employing deep learning for classification,

TABLE I
PERFORMANCE METRICS FOR DIFFERENT MODELS

Model	Accuracy	Precision	Recall	F1 Score
EfficientNetB7	84.00%	85.00%	83.00%	84.40%
ResNet50	74.00%	79.00%	73.00%	74.00%
VGGNet19	66.67%	67.45%	66.67%	67.00%
AlexNet	59.26%	58.88%	59.26%	58.71%



Fig. 5. Confusion matrix using EfficientNet-B7 Model

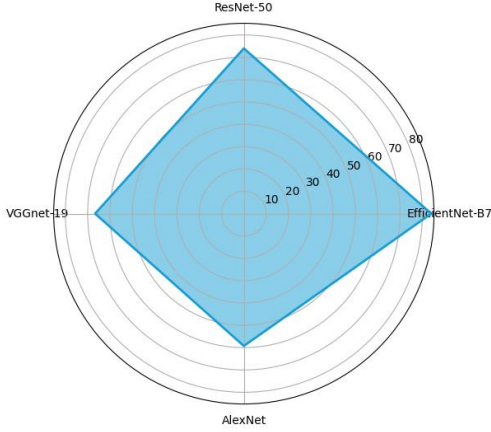


Fig. 6. Accuracy

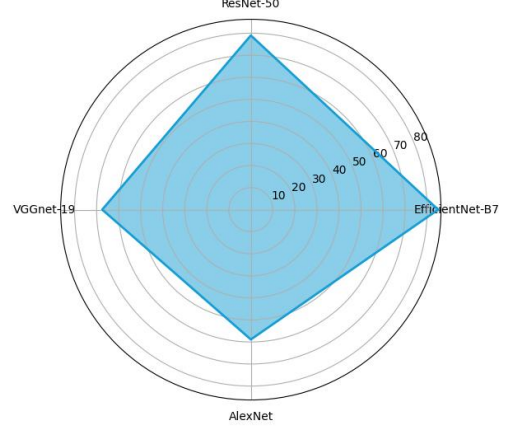


Fig. 7. Precision

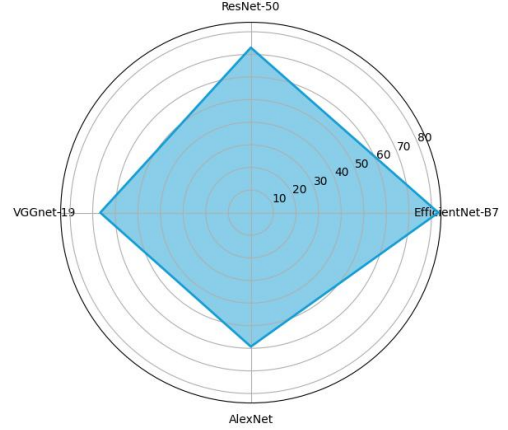


Fig. 8. Recall

this approach aids ophthalmologists in delivering more precise and data-supported diagnoses.

Future research might focus on improving classification models further, analyzing larger and more diverse datasets, and integrating explainability techniques to enhance clinical interpretability. These advancements will continue to refine AI-assisted ophthalmology, making automated strabismus detection more accessible and reliable.

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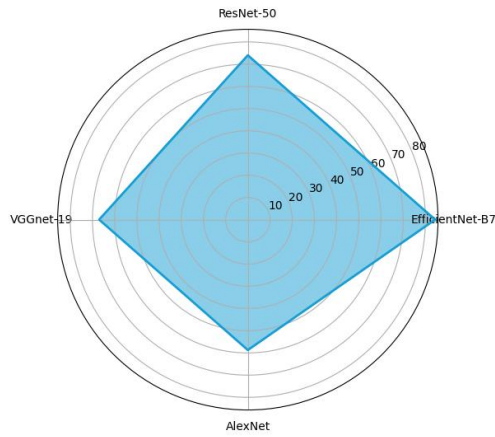


Fig. 9. F1-score

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